

Week 10: Tracking the Macroeconomy

(based on Chapter 11 of your text)

- **How to measures the economy** : Aggregated output and Income
- **Gross domestic product (GDP)** : The three ways of calculating GDP
- **Real GDP and Nominal GDP**
- **Price index and inflation rate**

“China passes Japan as Second-largest economy,”
2010 *New York Times*

GDP in US Billion Dollars, 2015

1	United States	17947.0
2	China	10866.4
3	Japan	4123.3
4	Germany	3355.8
5	United Kingdom	2848.8
6	France	2421.7
7	India	2073.5
8	Italy	1814.8
9	Brazil	1774.7
10	Canada	1550.5
11	Korea, Rep.	1377.9
12	Australia	1339.5
13	Russian Federation	1326.0
14	Spain	1199.1
15	Mexico	1144.3
16	Indonesia	861.9
17	Netherlands	752.5
18	Turkey	718.2
19	Switzerland	664.7
20	Saudi Arabia	646.0

World Bank Data

62 UZ: 66.7

Ranking by GDP and per capita GDP, 2019

1	United States	21,482.41	24.4	-	24,671	1	2.54	65,062	8	North America
2	China	14,172.20	16.1	7310	19,581	2	6.18	10,099	72	Asia
3	Japan	5,220.57	5.93	8952	5,908	3	0.94	41,418	26	Asia
4	Germany	4,117.07	4.67	1104	4,937	4	1.86	49,692	18	Europe
5	India	2,957.72	3.36	1159	4,330	5	7.44	2,188	145	Asia
6	France	2,844.70	3.23	113	3,364	6	1.62	43,500	21	Europe
7	United Kingdom	2,809.91	3.19	34.8	3,257	7	1.49	42,036	23	Europe
8	Italy	2,112.80	2.40	697	2,396	8	1.03	34,784	28	Europe
9	Brazil	1,929.71	2.19	183	2,351	9	2.37	9,160	75	South America
10	Canada	1,820.36	2.07	109	2,322	10	2.05	48,601	19	North America
11	Korea	1,699.68	1.93	121	2,055	11	2.58	32,766	30	Asia
12	Russia	1,649.21	1.87	50.5	1,818	12	1.80	11,461	68	Europe
13	Spain	1,474.12	1.67	175	1,758	14	2.20	31,906	33	Europe
14	Australia	1,464.41	1.66	9.71	1,794	13	2.77	57,204	11	Oceania

World Bank Data : Country by GDP

<http://data.worldbank.org/indicator/NY.GDP.MKTP.CD>

GDP Growth (annual %)

GDP per capita (current US\$)

GDP per capita Growth (annual %)

GNI per capita, PPP (current international \$)

Inflation, GDP deflator (annual %)

* PPP: Purchasing Power Parity

GDP per Capita 2015

1	Luxembourg	101450	16	Canada	43249
2	Switzerland	80215	17	Hong Kong SAR, China	42423
3	Macao SAR, China	78586	18	Finland	41921
4	Norway	74735	19	Germany	41219
5	Qatar	74667	20	United Arab Emirates	40438
6	Australia	56328	21	Belgium	40231
7	United States	55837	22	New Zealand	37808
8	Singapore	52889	23	Brunei Darussalam	36608
9	Denmark	52002	24	France	36248
10	Ireland	51290	25	Israel	35330
11	Sweden	50273	26	Japan	32477
12	Iceland	50173	27	Italy	29847
13	Netherlands	44433	28	Kuwait	28985
14	United Kingdom	43734	29	Korea, Rep.	27222
15	Austria	43439	30	Spain	25832

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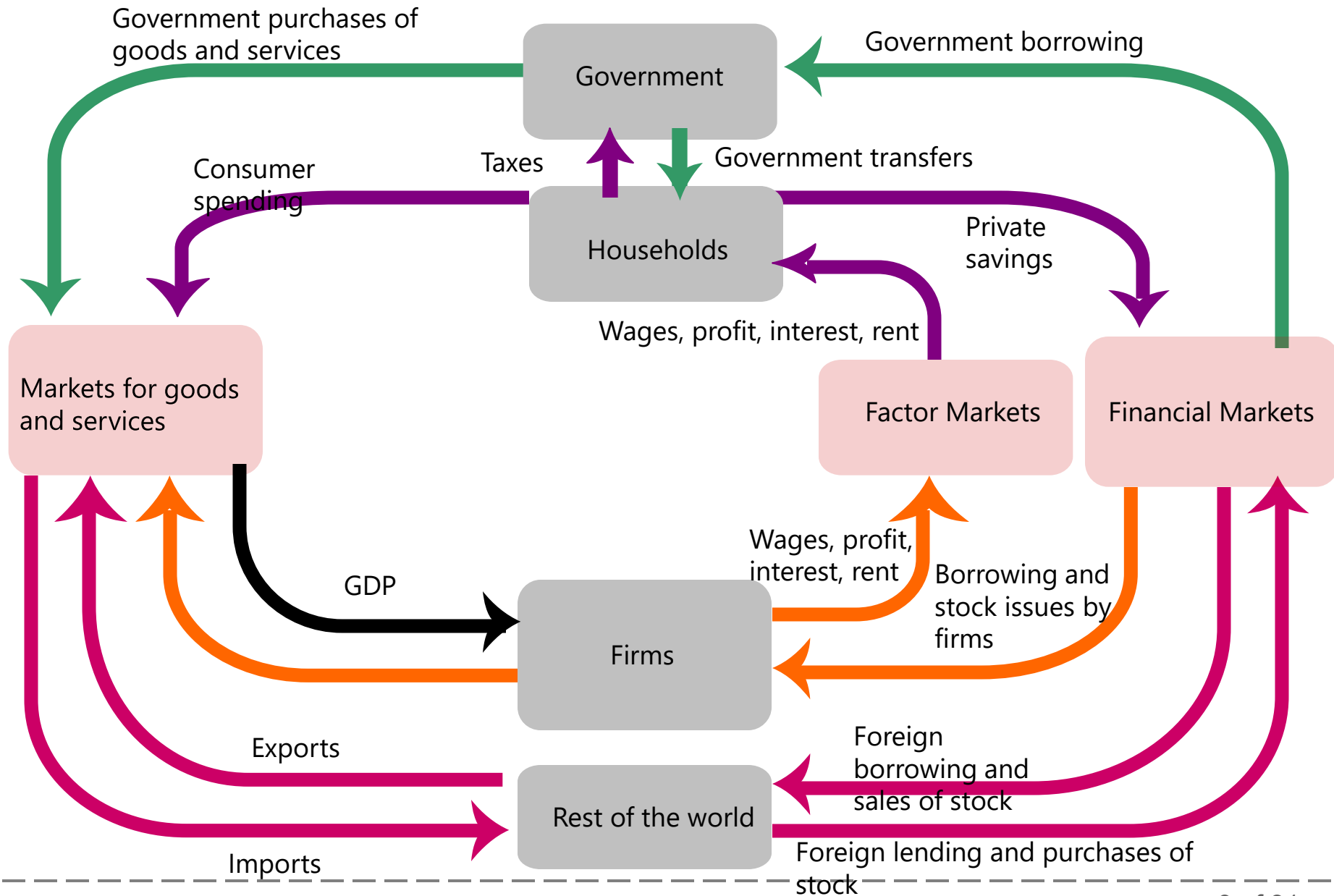
Popular World Development Indicators (World Bank)

GDP growth (annual %)
GDP (current US\$)
GDP per capita (current US\$)
GNI per capita, Atlas method (current US\$)
Foreign direct investment, net (BoP, current US\$)
GNI per capita, PPP (current international \$)
GINI index (World Bank estimate)
Inflation, consumer prices (annual %)
Inflation, GDP deflator (annual %)
Internet users (per 100 people)
Life expectancy at birth, total (years)
Literacy rate, adult total (% of people ages 15 and above)
Unemployment, total (% of total labor force) (modeled ILO estimate)
Poverty headcount ratio at national poverty lines (% of population)
Agriculture, value added (% of GDP)
CO2 emissions (metric tons per capita)
Central government debt, total (% of GDP)
Population, total

The National Accounts

- Almost all countries calculate a set of numbers known as the ***national income and product accounts (NIPA)***.
- The national income and product accounts, or national accounts, keep track of the flows of money between different parts of the economy.

An Expanded Circular-Flow Diagram



Income: Where does it come from?

- Households earn income via the factor markets from **wages**, interest on **bonds**, **dividends** on **stocks**, and **rent** on land.
- A **stock** is a share in the ownership of a company held by a shareholder.
- A **bond** is borrowing in the form of an IOU that pays interest.
- In addition, households receive **government transfers** from the government.
- **Disposable income**, total household income minus taxes, is available to spend on consumption or to save.

The National Accounts

- **Private savings** : disposable income that is not spent on consumption
= disposable income - consumer spending
- The banking, stock, and bond markets, which channel private savings and foreign lending into investment spending, government borrowing, and foreign borrowing, are known as the **financial markets**.

The National Accounts

- ***Government purchases of goods and services*** (G) is paid for by tax receipts as well as by ***government borrowing***.
- ***Exports*** (X) generate an inflow of funds into the country from the rest of the world, while ***imports*** (M) lead to an outflow of funds to the rest of the world.

The National Accounts

- Inventories are stocks of goods and raw materials held to facilitate business operations.
- Investment spending is spending on productive physical capital, such as machinery and construction of structures, and on changes to inventories.
- Final goods and services are goods and services sold to the final, or end, user.
- Intermediate goods and services are goods and services—bought from one firm by another firm—that are inputs for production of final goods and services.

Gross Domestic Product

- **Gross domestic product** or **GDP** measures the **total value of all final goods and services** produced in the economy during a given year. It does not include the value of *intermediate goods*.
- **Aggregate spending**, the sum of consumer spending, investment spending, government purchases of goods and services, and exports minus imports, the total spending on domestically produced final goods and services in the economy.

$$C + I + G + X - M$$

Three ways to calculate GDP

- (1) Add up the ***value added*** of all producers
- (2) Add up all spending on domestically-produced final goods and services. This results in the equation: **GDP = C + I + G + X - M**
- (3) Add up all income paid to factors of production

Simon Kuznet calculated the first national income accounts

Calculating Gross Domestic Product

Total spending on domestically produced final goods and services = \$21,500

	American Ore, Inc.	American Steel, Inc.	American Motors, Inc.	Total factor income
Value of sales	\$4,200 (ore)	\$9,000 (steel)	\$21,500 (car)	
Intermediate goods	0	4,200 (iron ore)	9,000 (steel)	
Wages	2,000	3,700	10,000	\$15,700
Interest payments	1,000	600	1,000	2,600
Rent	200	300	500	1,000
Profit	1,000	200	1,000	2,200
Total expenditure by firm	4,200	9,000	21,500	
Value added per firm	4,200	4,800	12,500	
=				
Value of sales – cost of intermediate goods				

Total payments to factors = \$21,500

Sum of value added = \$21,500

Our Imputed Lives

- Some economists have produced alternative measures that try to “impute” the value of household. But the standard measure of GDP doesn’t contain that imputation.
- GDP estimates do, however, include an imputation for the value of “owner-occupied housing.” If you buy the home you were formerly renting, GDP does not go down. Statisticians make an estimate of what you would have paid if you rented whatever you live in, whether it’s an apartment or a house.
- To be accurate, estimates of GDP must take into account the value of housing that is occupied by owners as well as the value of rental housing.

GDP: What's In and What's Out

Included

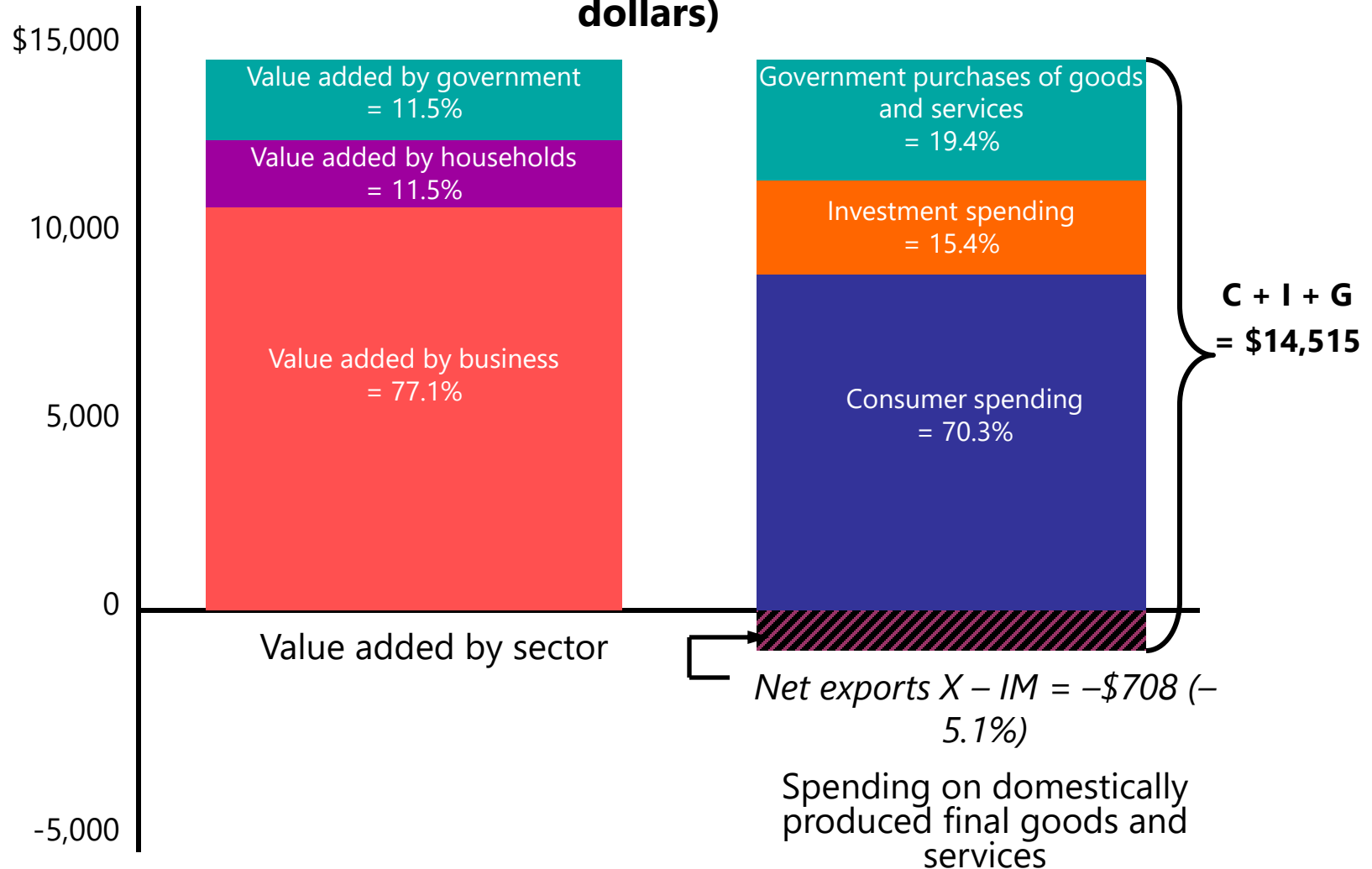
- domestically produced final goods and services
- new construction of structures
- changes to inventories

Not Included

- intermediate goods and services
- inputs
- used goods
- **Household works**
- financial assets like stocks and bonds
- foreign-produced goods and services

Calculating Gross Domestic Product

Components of GDP (billions of dollars)



Creating the National Accounts

- The national accounts owe their creation to the Great Depression. All government officials had were scattered statistics: railroad freight car loadings, stock prices, and incomplete indexes of industrial production.
- Simon Kuznets developed a set of national income accounts. The first version of these accounts was presented to Congress in 1937 and in a research report titled *National Income*.
- The push to complete the national accounts came during World War II, when policy makers were in even more need of comprehensive measures of the economy's performance. The federal government began issuing estimates of gross domestic product and gross national product in 1942.

Real vs. Nominal GDP

- ***Real GDP*** is the total value of the final goods and services produced in the economy during a given year, calculated using the prices of a selected base year.
- ***Nominal GDP*** is the value of all final goods and services produced in the economy during a given year, calculated using the prices current in the year in which the output is produced.

Real vs. Nominal GDP

Nominal GDP and Real GDP					
	2014		2015		
	Price	Quantity	Price	Quantity	
Apples	5	100	6	110	
Computers	1000	50	1100	60	
Nominal GDP	= 5 * 100 + 1000 * 50		= 6 * 110 + 1100 * 60		
	50500		66660		
Real GDP in 2014 Price	= 5 * 100 + 1000 * 50		= 5 * 110 + 1000 * 60		
	50500		60550		
Real GDP in 2015 Price	= 6 * 100 + 1100 * 50		= 6 * 110 + 1100 * 60		
	55600		66660		

$$\sum P_0 * Q_0, \quad \sum P_1 * Q_1, \quad \sum P_0 * Q_1, \quad \sum P_1 * Q_0$$

Real vs. Nominal GDP

- ***Chained dollars*** is the method of calculating changes in real GDP using the average between the growth rate calculated using an early base year and the growth rate calculated using a late base year.
- ***GDP per capita*** is a measure of average GDP per person, but is not by itself an appropriate policy goal.

Real vs. Nominal GDP

Nominal versus Real GDP in 1993, 2000, and 2007

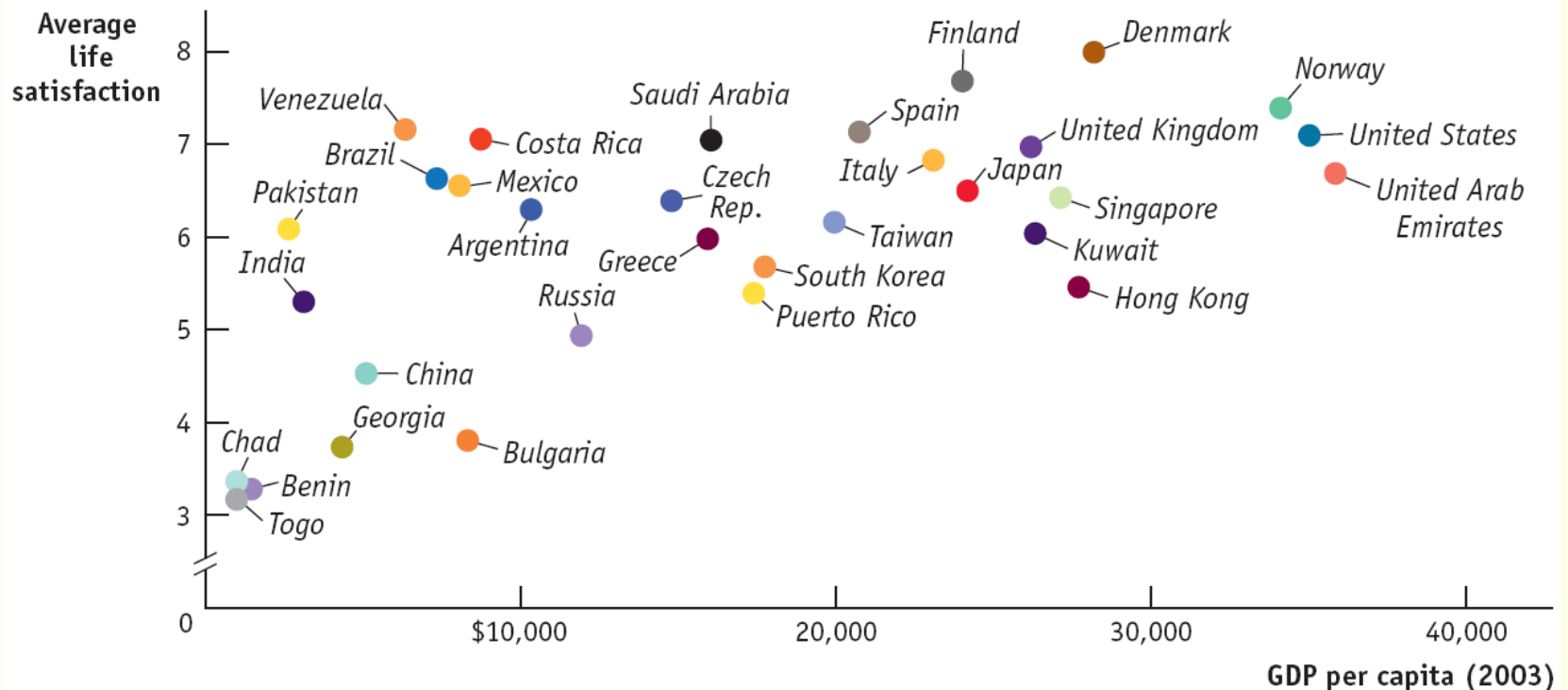
	Nominal GDP (billions of current dollars)	Real GDP (billions of 2000 dollars)
1993	\$6,657	\$7,533
2000	9,817	9,817
2007	13,808	11,524

GDP deflator of year 2007 in year 2000 price:
 $= 13,808 / 11,524 * 100 = 119.8$

GLOBAL COMPARISON

GDP and the meaning of life

- Rich is better
- Money matters less as you grow richer
- Money isn't everything

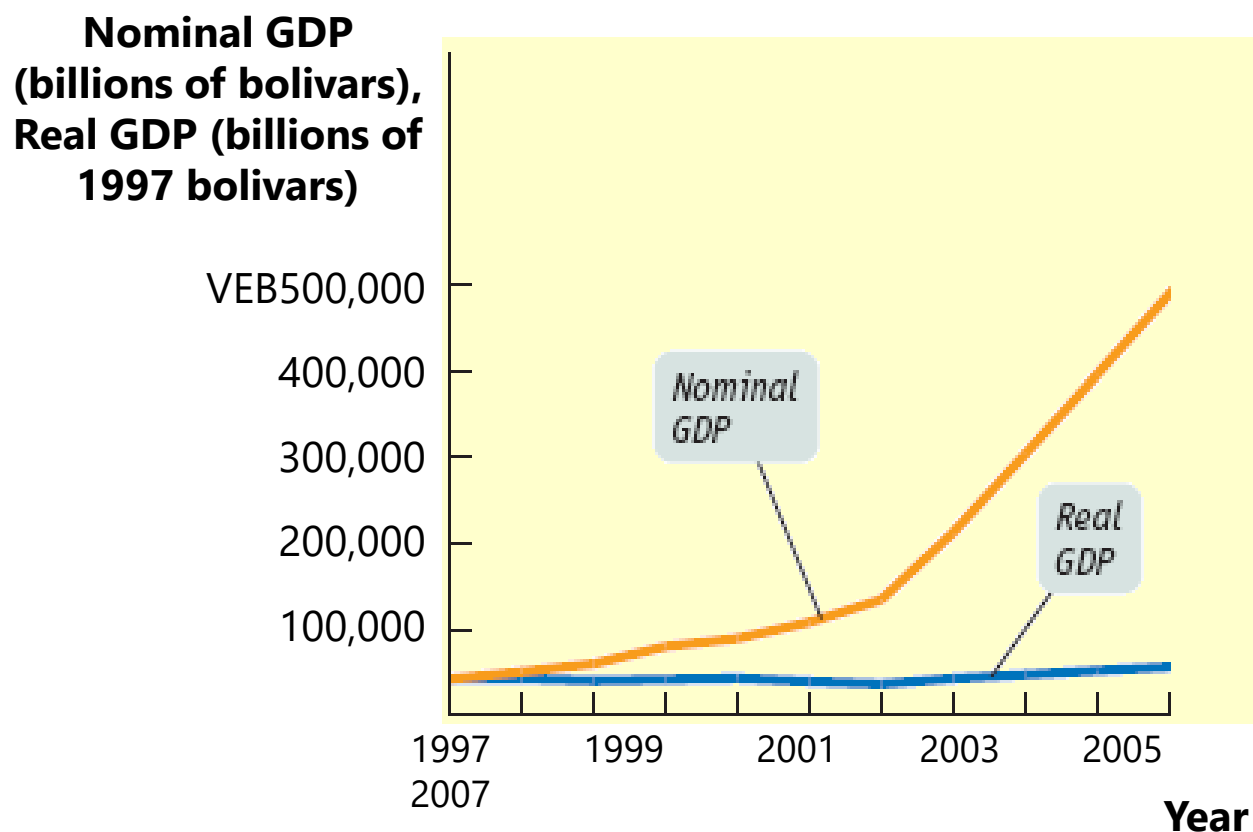


Miracle in Venezuela?

- The South American nation of Venezuela has a distinction that may surprise you: in recent years, it has had one of the world's fastest-growing nominal GDPs. Between 1997 and 2007, Venezuelan nominal GDP grew by an average of 28% each year—much faster than nominal GDP in the United States or even in booming economies like China.
- So is Venezuela experiencing an economic miracle?

Miracle in Venezuela?

- No, it's just suffering from unusually high inflation.



Price Indexes and the Aggregate Price Level

- The **aggregate price level** is a measure of the overall level of prices in the economy.
- To measure the aggregate price level, economists calculate the cost of purchasing a **market basket**.
- A **price index** is the ratio of the current cost of that market basket to the cost in a base year, multiplied by 100.

$$\text{Price index in a given year} = \frac{(\text{Cost of market basket in a given year})}{(\text{Cost of market basket in base year})} \times 100$$

Market Baskets and Price Indexes

Calculating GDP and Real GDP in a Simple Economy

	Pre-frost	Post-frost
Price of orange	\$0.20	\$0.40
Price of grapefruit	0.60	1.00
Price of lemon	0.25	0.45
Cost of market basket (200 oranges, 50 grapefruit, 100 lemons)	$(200 \times \$0.20) +$ $(50 \times \$0.60) +$ $(100 \times \$0.25) = \95.00	$(200 \times \$0.40) +$ $(50 \times \$1.00) +$ $(100 \times \$0.45) = \175.00

Price index: pre-frost = 100

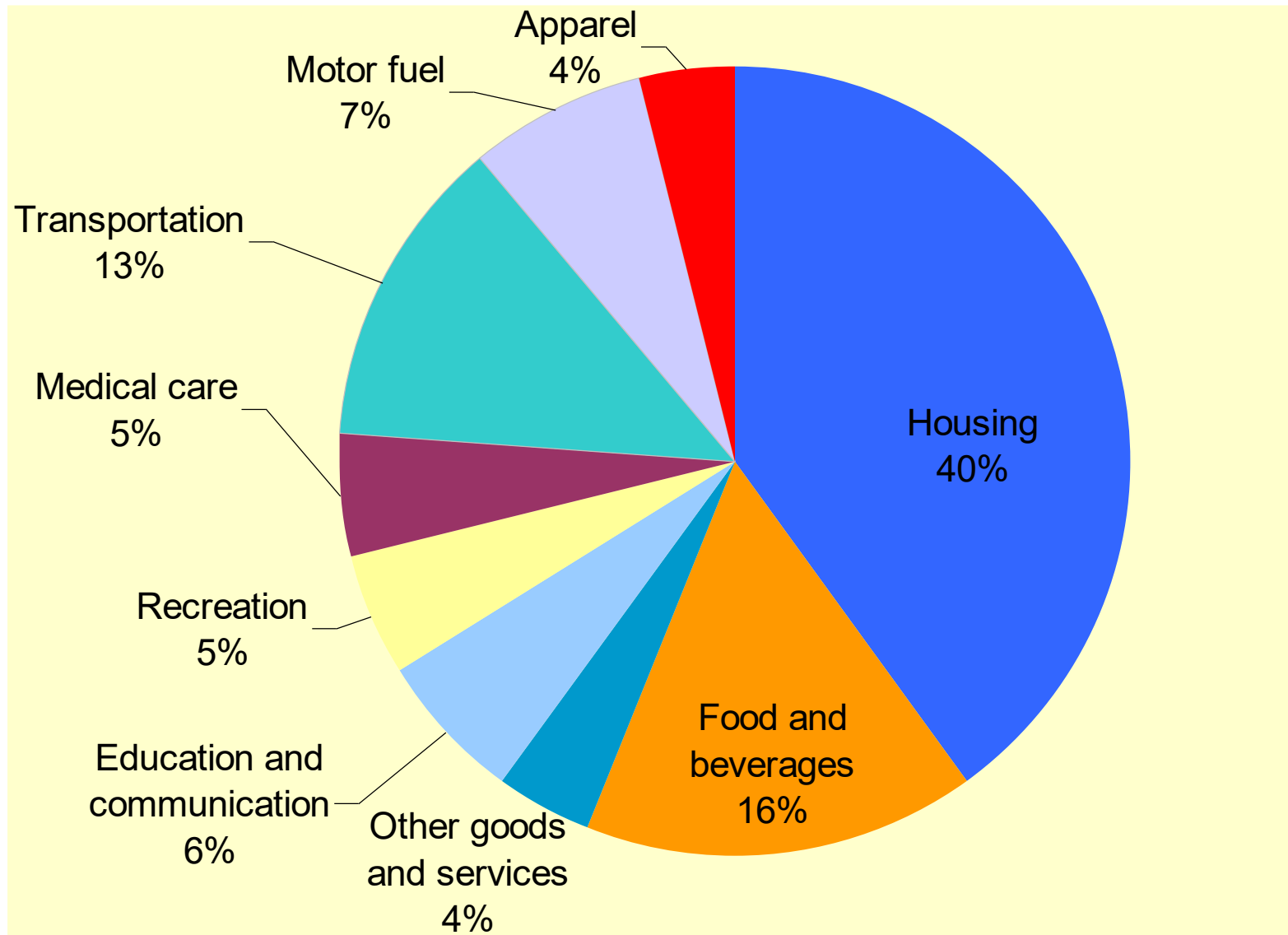
post-frost = $175 / 95 * 100 = 184$

Inflation Rate, CPI, and other Indexes

- The ***inflation rate*** is the yearly percentage change in a price index, typically based upon ***Consumer Price Index***, or ***CPI***, the most common measure of the aggregate price level.
- The ***consumer price index***, or ***CPI***, measures the cost of the market basket of a typical urban American family.

$$\text{Inflation rate} = \frac{(\text{Price index in year 2} - \text{Price index in year 1})}{(\text{Price index in year 1})} \times 100$$

Consumer Price Index



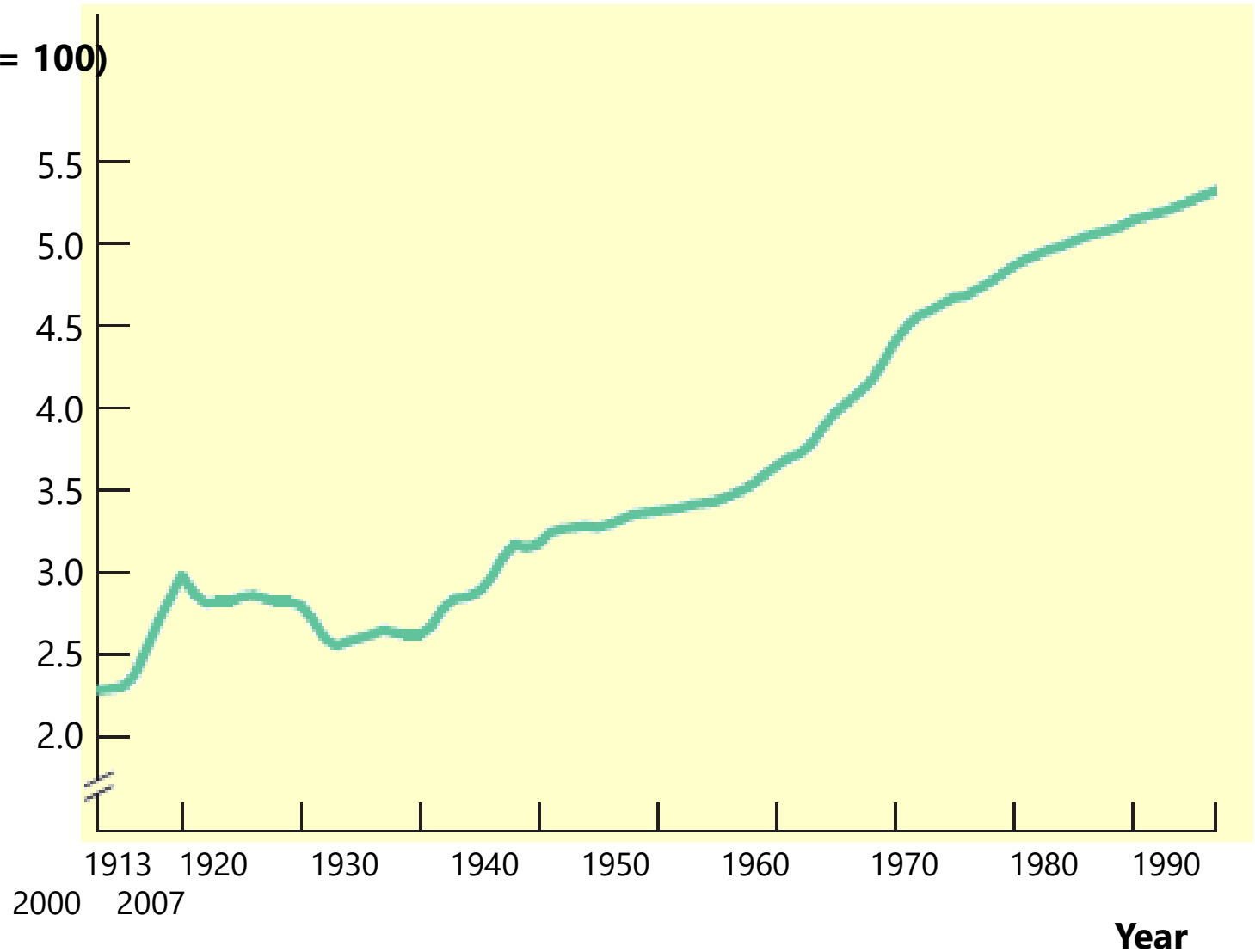
Is the CPI biased?

- The U.S. government takes considerable care in measuring consumer prices. Nonetheless, many economists believe that the consumer price index systematically **overstates** the actual rate of inflation.
- One reason is the fact that the CPI measures the cost of buying a given market basket. Yet, consumers typically alter the mix of goods and services they buy, reducing purchases of products that have become relatively more expensive and increasing purchases of products that have become relatively cheaper.
- The second reason arises from innovation. By widening the range of consumer choice, innovation makes a given amount of money worth more.

Consumer Price Index

Log CPI

(1982 - 1984 = 100)

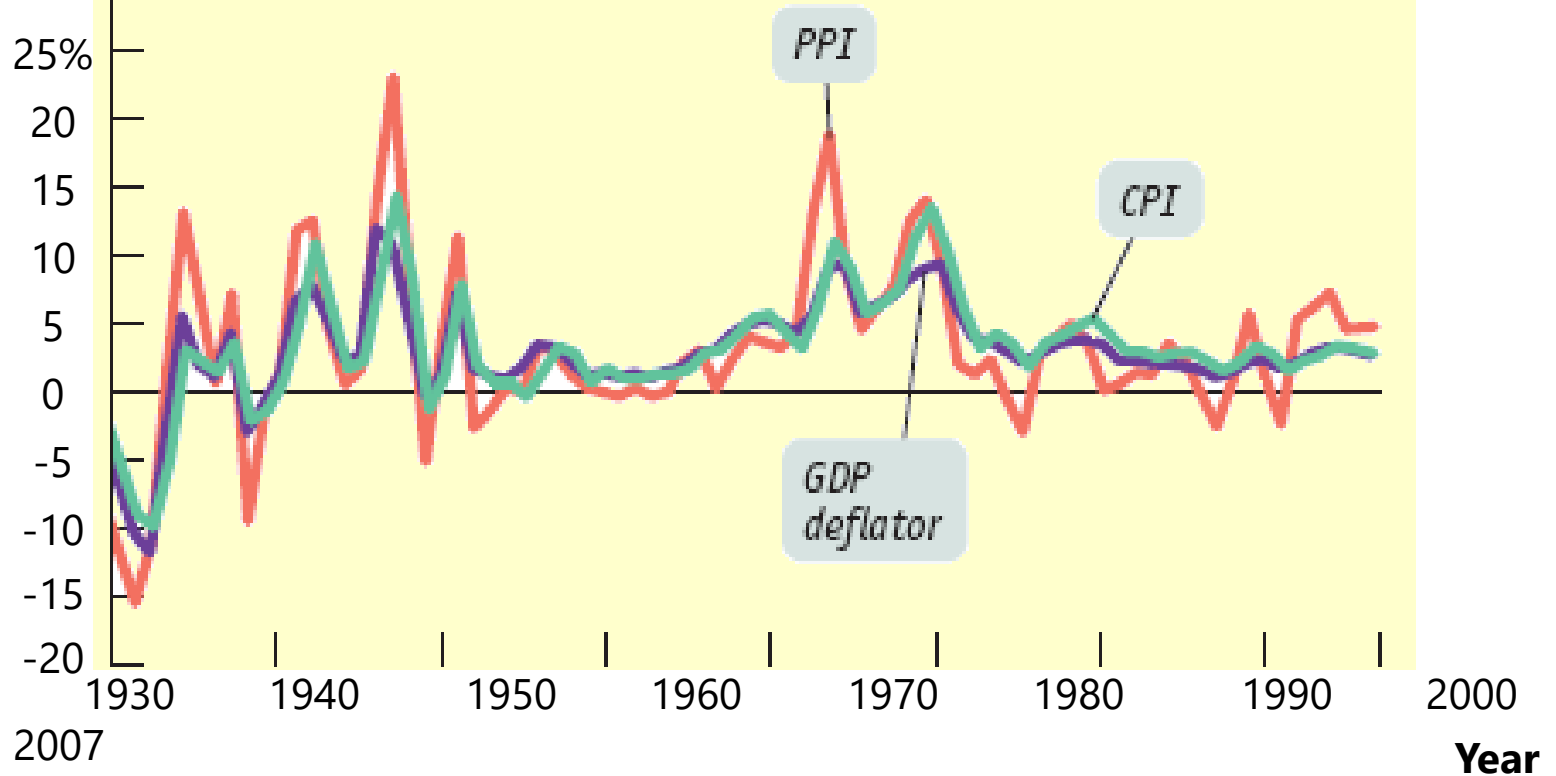


Other Price Measures

- A similar index to CPI for goods purchased by firms is the ***producer price index***.
- Economists also use the ***GDP deflator***, which measures the price level by calculating the ratio of nominal to real GDP.

The CPI, the PPI, and the GDP Deflator

Percent change in
CPI, PPI, GDP
deflator



Indexing to the CPI

- The CPI has a direct and immediate impact on millions of Americans. The reason is that many payments are tied, or “indexed,” to the CPI—the amount paid rises or falls when the CPI rises or falls.
- Today, 48 million people receive checks from Social Security. The amount of an individual’s check is determined by a formula that reflects his or her previous payments into the system as well as other factors. In addition, all Social Security payments are adjusted each year to offset any increase in consumer prices over the previous year. The CPI is used to calculate the official estimate of the inflation rate used to adjust these payments yearly.

1. Economists keep track of the flows of money between sectors with the **national income and product accounts**, or **national accounts**. Households earn income via the factor markets from wages. **Disposable income** is allocated to **consumer spending** (C) and **private savings**. Via the **financial markets**, private savings and foreign lending are channeled to **investment spending** (I), government borrowing, and foreign borrowing. **Government purchases of goods and services** (G) are paid for by tax revenues and any **government borrowing**. **Exports** (X) generate an inflow of funds into the country from the rest of the world, but **imports** (IM) lead to an outflow of funds to the rest of the world.

2. **Gross domestic product, or GDP**, measures the value of all **final goods and services** produced in the economy. It does not include the value of **intermediate goods and services**, but it does include **inventories** and **net exports** ($X - M$). It can be calculated in three ways: add up the **value added** by all producers; add up all spending on domestically produced final goods and services ($GDP = C + I + G + X - M$); or add up all the income paid by domestic firms to factors of production. These three methods are equivalent.

3. **Real GDP** is the value of the final goods and services produced calculated using the prices of a selected base year. Except in the base year, real GDP is not the same as **nominal GDP**, the value of **aggregate output** calculated using current prices. Analysis of the growth rate of aggregate output must use real GDP. Real **GDP per capita** is a measure of average aggregate output per person but is not in itself an appropriate policy goal. U.S. statistics on real GDP are always expressed in **chained dollars**.

4. To measure the **aggregate price level**, economists calculate the cost of purchasing a **market basket**. A **price index** is the ratio of the current cost of that market basket to the cost in a selected base year, multiplied by 100.
5. The **inflation rate** is the yearly percent change in a price index, typically based on the **consumer price index**, or **CPI**, the most common measure of the aggregate price level. A similar index for goods and services purchased by firms is the **producer price index**, or **PPI**. Finally, economists also use the **GDP deflator**.

- Answer Questions 1-3, Chapter 10, p.324.